COMPUTER VISION · PROJECT OVERVIEW

# Intel Image Classification with CNNs

MobileNetV2 transfer learning pipeline for classifying natural and urban scenes.



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01 · PROJECT PURPOSE

# What the project solves

**Automate scene recognition** instead of relying on slow, inconsistent manual classification.

**Predict one of six categories:** buildings, forest, glacier, mountain, sea, or street.

**Deliver reproducible outputs:** evaluation, visualizations, and per-image predictions.

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02 · DATASET

# The dataset covers six scene categories



**Buildings**

Built-up areas



**Forest**

Forested landscapes



**Glacier**

Ice-covered landscapes



**Mountain**

Mountainous landscapes



**Sea**

Coastal water scenes



**Street**

Roads and urban scenes

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# The workflow turns images into predictions

01

**Download**

Download and verify the Kaggle dataset with KaggleHub.

02

**Prepare**

Resize RGB images to 160 × 160 and batch them in groups of 32.

03

**Split**

Split the provided training directory into 80% training and 20% validation.

04

**Evaluate**

Keep the provided test directory separate for final evaluation.

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# Augmentation improves training variety



### Flip

Random horizontal flips.



### Rotate

Small rotations.



### Zoom

Random zoom during training.

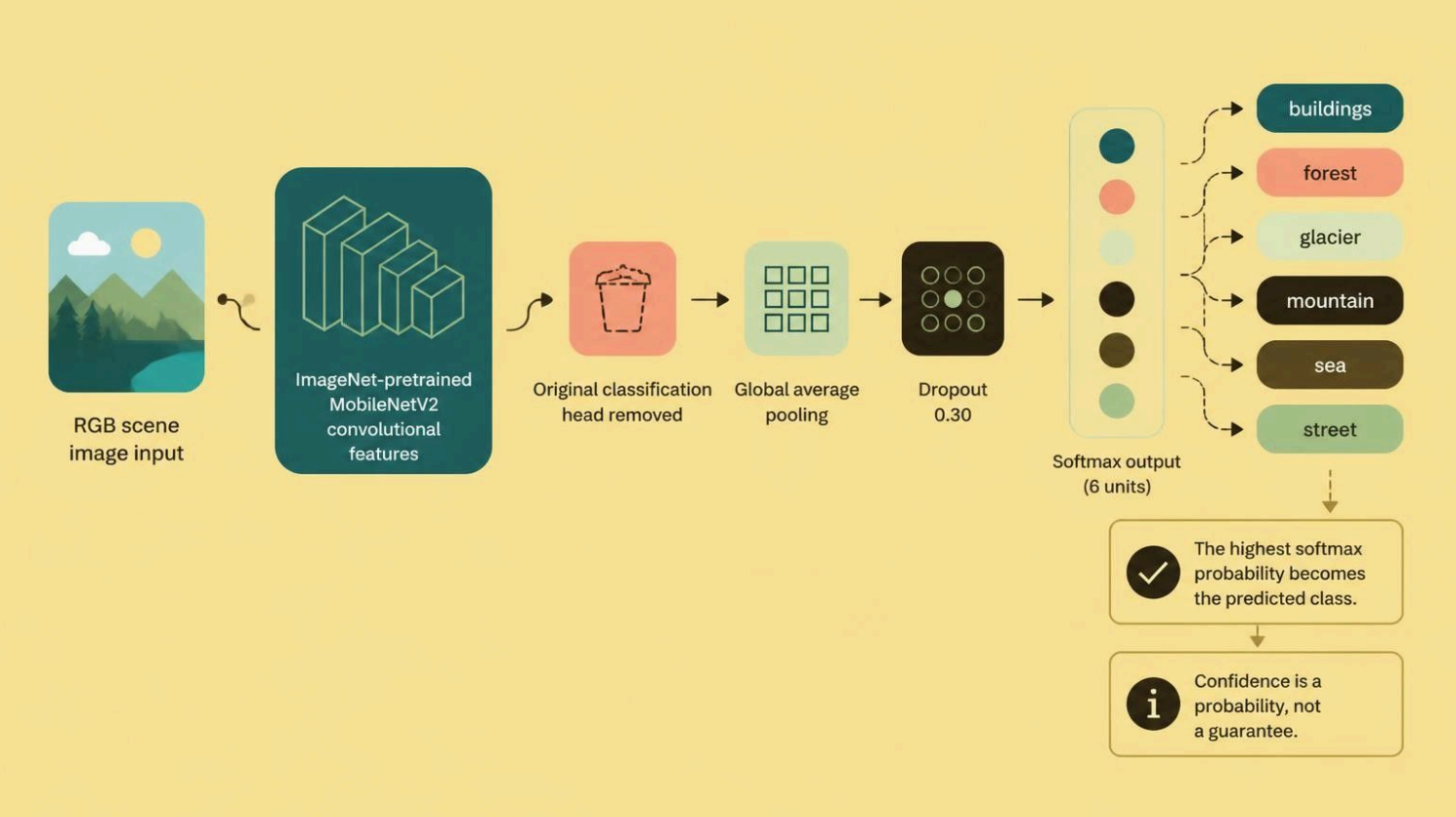
### Training variety, unchanged task.

Keep held-out test images unchanged. Augmentation exposes the model to modest visual variations without changing the task.

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05 · MODEL ARCHITECTURE

# MobileNetV2 supplies the visual backbone




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# Training happens in two phases

01

### Phase 1 · Train the head

Freeze the pretrained feature extractor and train the new head for **8 epochs**.

02

### Phase 2 · Fine-tune

Unfreeze the last **30 base-model layers**, keep batch-normalization layers frozen, and fine-tune for up to **7 additional epochs** at a learning rate of **0.00001**.

Save the checkpoint with the best validation accuracy. Reduce learning rate when validation loss plateaus.

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07 · EVALUATION

# The model is evaluated beyond accuracy

### Accuracy

Share of correct test predictions.

### Precision · recall · F1-score

Class-level behavior.

### Confusion matrix

Which scene categories are being confused.

### Test loss

Gap between predicted probabilities and true labels.

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# Results must come from the completed notebook

CELL 12

## Test accuracy

[enter value printed by Cell 12]

Test loss: [enter value printed by Cell 12]

CELL 13

## Classification report

Use Cell 13 for per-class precision, recall, and F1-score.

No numerical performance claim is made here because the documented run has not been executed. Exact metrics depend on the downloaded dataset version and completed notebook run.

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09 · NEXT STEPS

# The project is ready for analysis and extension

**Export the trained model, learning curves, confusion matrix, sample predictions, and prediction CSV.**

**Compare MobileNetV2 with a custom CNN or other pretrained architectures.**

**Extend the analysis** with class-wise error analysis, hyperparameter tuning, and a small image-upload inference interface.

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